

Advanced Analysis: The Hydrological Cycle

4. Mastery of Key Terminology

To accurately describe the hydrological cycle, you must be precise with your scientific vocabulary. Review the terms in the word bank and match them to their correct definitions based on our study of energy transfer and closed systems.

Word bank: Condensation, Evaporation, Precipitation, Solar Energy, Closed System, Climate Regulation

Definition	Word
1. The electromagnetic radiation emitted by the sun that acts as the primary engine driving the water cycle.	
2. A physical system that does not allow certain types of matter to enter or leave; explains why the Earth's total water mass remains constant.	
3. The process where liquid water absorbs energy to change state into gaseous water vapour.	
4. The phase change from water vapour back into liquid droplets, occurring when moisture-laden air cools.	
5. Any form of water—liquid or solid—that falls from the atmosphere to the Earth's surface (e.g., rain, snow, hail).	
6. The movement of heat via water vapour and ocean currents from the equator toward the poles to balance global temperatures.	

2. Systems Thinking: The Closed System Paradox

The total mass of water on Earth remains constant; it is a "closed system" where matter is neither created nor destroyed. However, the *distribution* of this water changes constantly.

Question: If the total amount of water on Earth is fixed, explain why water scarcity is an increasing global crisis. In your answer, distinguish between the concepts of *total water mass* versus *available freshwater resources*.

.....

.....

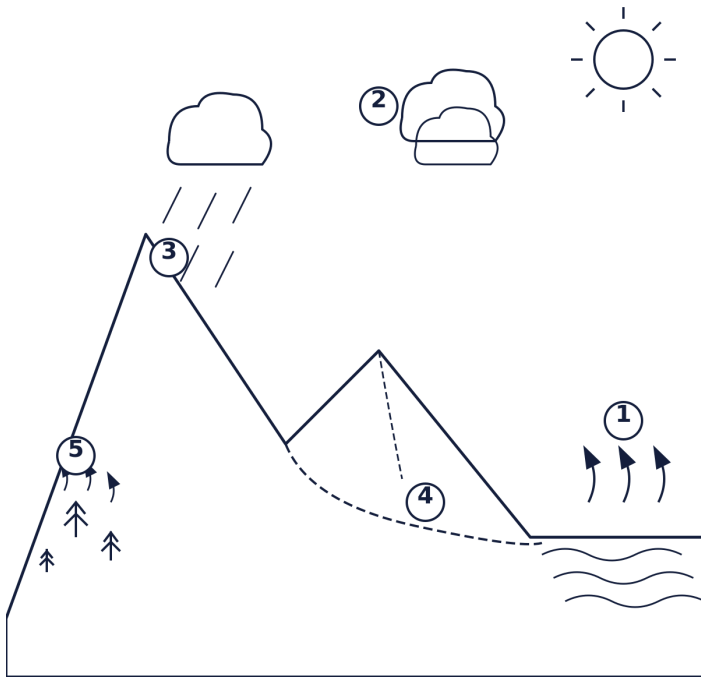
.....

.....



1. The Hydrological Cycle: Diagram & Analysis ☁️

Examine the diagram below. Beyond simply identifying the stages, analyse the physical state changes involved in each step of the cycle.



#	Process Term	State Change (e.g., Liquid → Gas)
1		
2		
3		
4		
5		

3. Extension: Thermodynamics & Climate



Latent Heat: The energy absorbed or released by a substance during a change in its physical state (phase) that occurs without changing its temperature.

The water cycle does more than just move water; it moves **energy**. Using the concept of *latent heat*, explain how the water cycle regulates global temperatures. specifically addressing how energy is transported from the equator toward the poles.
